When should a neural text-entry decoder commit? Risk-controlled commit decisions from a decoder-agnostic trust layer, evaluated on intracortical, EMG and MEG data

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Abstract

Objective. Neural text-entry decoders increasingly end in a language model that turns weak neural evidence into fluent sentences, but they rarely signal when a sentence is wrong. We ask when such a system should commit a sentence without confirmation, and how to bound how often it commits wrong text. *Approach.* We define an open per-frame output format (NeuralTextStream) and a trust layer that reads only this stream, calibrates confidence and chooses to commit, offer alternatives or abstain, with the commit threshold set by conformal risk control, Learn-then-Test or adaptive online updates. We re-fit one implementation offline on public data from one intracortical speech participant (816 test trials), forearm EMG key presses composed into sentences (19 participants) and MEG typing (19 participants). The primary outcome is the wrong-commit rate per sentence attempted, with the commit rate and the error rate among commits; a simulated gaze-typing user gives effective words per minute (eWPM). *Main results.* On intracortical speech, the layer cut wrong commits from 61% to 5.9% of sentences (our decoder) and from 19% to 4.4% (published decoder); 25% and 9% of committed sentences were still wrong. It was 0.18 and 0.66 eWPM faster than always offering (95% CIs over days [0.12, 0.24], [0.53, 0.80]), but under most user-model variants not faster than always committing. On EMG and MEG it committed at most 4% and 0% of sentences, and gaze typing with word completion was faster on MEG and the two weaker EMG decoders. With calibration data disjoint from all fitted parts, offline conformal calibration gave 2.5–6.0% on later data (target 5%), never significantly above the rate for exchangeable data; an apparent drift in our earlier analysis came from reusing fit data. The layer committed no text from signal-destroying control streams but cannot detect a real stream of a different intended sentence. *Significance.* A stream-level commit rule makes the commit decision explicit and bounds how often wrong text is sent, at little or no speed cost relative to always offering. Speed gains are secondary, user-model dependent and need a user study.

1 Introduction

Brain–computer interfaces (BCIs) for text entry are moving towards clinical and consumer use. Intracortical neuroprostheses decode attempted speech, handwriting and typing at high rates [1–4]; non-invasive decoders reconstruct typed sentences from magnetoencephalography (MEG) [5, 6]; and surface electromyography (sEMG) is a candidate input for wearables [7–9]. Most of these systems end in a language-model (LM) stage that turns noisy per-frame predictions into sentences. This lowers the error rate but changes the kind of error: when the neural evidence is weak, the LM still returns a grammatical sentence, and a reader has no cue that it is wrong. In speech recognition, such fluent unsupported output has caused documented harm [10]. For a person who depends on the interface to communicate, a wrong sentence sent in their name can cost more than a slow one.

BCI spellers handle errors inside one system: P300 spellers stop when a character's posterior passes a threshold [11, 12], error-related potentials flag wrong selections [13], and myoelectric

control rejects uncertain decisions [14]. Sentence-level LM decoders for speech, handwriting, typing and sEMG now share one structure, per-frame probabilities followed by an LM search, which makes a decoder-independent commit rule possible: a rule that decides whether a sentence is committed (sent without confirmation), offered for confirmation or not shown, and that bounds how often it commits wrong text. Three things are missing. There is no common output format: the Apple BCI HID profile sends events with a "neural activity strength" that is not a probability [15], W3C EMMA has no per-frame distributions [16], and Lab Streaming Layer and BIDS carry raw signals [17, 18]. Calibration, selective prediction and conformal risk control are well studied [19–21] and have been applied to single BCI decoders [22, 23] and to LM correction in speech recognition [24], but to our knowledge not combined into a commit rule for neural text entry with a stated guarantee and tested on more than one modality. And the fair comparator for a commit rule, "always offer, never commit", is rarely reported next to a no-BCI baseline such as gaze typing [25].

This work. We define the problem (Section 3), a per-frame output format (Section 4), a trust layer that reads only this format (Section 5) and an evaluation protocol (Section 6), and we test them offline on public data from three modalities. Our contributions are:

1. **A decoder-agnostic trust layer** that reads only per-frame log-probabilities: temperature calibration, an optional neural-evidence guard and a COMMIT/OFFER/ABSTAIN policy, refitted per device. With a light reference decoder, it also gates the published sentence of a decoder whose weights we do not have.
2. **Wrong-commit control with stated guarantees** from conformal risk control and Learn-then-Test [21, 26] or adaptive online updates [27, 28], and evidence that their separate-data requirement matters: calibrating on the data that fitted the other parts produced an apparent drift failure that disappeared with clean calibration data.
3. **A safety-first evaluation protocol**: the wrong-commit rate with the commit rate and the error among commits; always-commit, always-offer, plain-threshold, gaze-typing and word-completion baselines; a simulated-user speed metric with a sensitivity analysis; and signal-destroying and intent-mismatch controls.
4. **An open format and release**: the NeuralTextStream v0.1 specification, adapters for three device types, a hash-chained audit log and all scripts under Apache-2.0.

Figure 1 summarises the result. The commit decision was tested non-trivially on one intracortical participant, where the layer turned frequent wrong commits into rare ones; on sEMG it committed at most 4% of composed sentences, and on MEG it correctly never committed (a negative control). Its speed gains were small and user-model dependent, so our claim is about safety: the layer makes the commit decision explicit and bounds how often wrong text is sent, at little or no speed cost relative to always offering.

2 Related work

Brain-to-text decoders. MEG decoding of typed sentences reached a CER of 0.29 with a keystroke-locked (synchronous) model [5]; a second version decodes without key times (asynchronous) and adds an LLM stage [6]. Other non-invasive work decodes perceived speech or words [29–31]. Intracortical decoders reach high accuracy for handwriting, speech and typing [1–4, 32], with public phoneme logits [33, 34]; sEMG text entry exists for the wrist and forearm [7–9]. Most use CTC [35] with beam search and an LM [36]. In the intracortical study whose data we use, decoded words were displayed and completed sentences read aloud by text-to-speech [2]; our layer acts at this point. **[TODO: R1: check in the supplement of Card et**

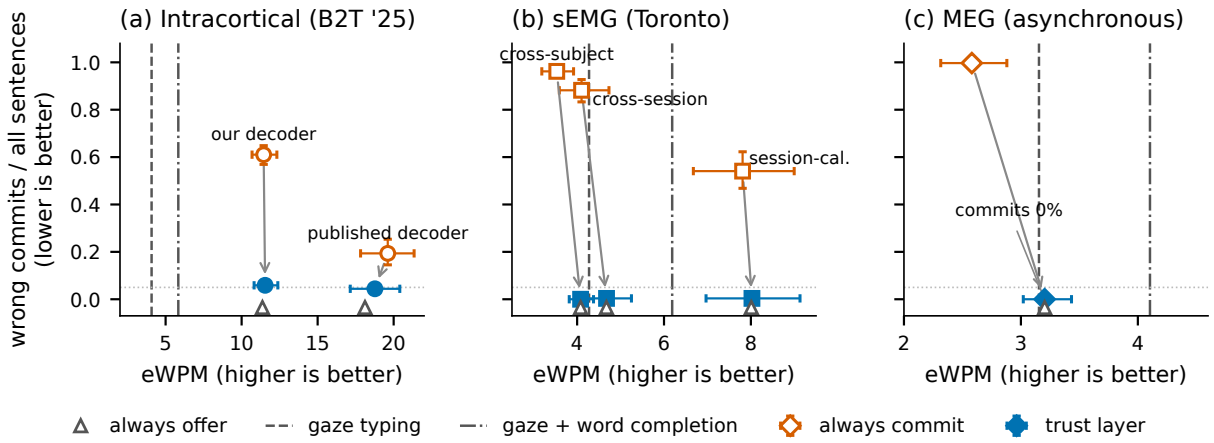


Figure 1: The trust layer turns frequent wrong commits into rare ones; on MEG it does so by never committing. Wrong commits per sentence attempted against effective words per minute (eWPM) of a simulated user. Open markers: always commit the top sentence; filled markers: the trust layer on the same output (arrows); triangles: always offer (no wrong commits; drawn just below 0). Dashed line: gaze typing; dash-dotted line: gaze typing with word completion; dotted line: $\alpha = 0.05$. Error bars: 95% cluster-bootstrap CIs over days (a), participants (b) and sentences × participants (c). (a) Intracortical speech (Brain-to-Text '25, one participant, 816 test trials): our phoneme-to-word decoder and the published sentence. (b) Forearm sEMG, 551 composed sentences, 19 participants, three decoder conditions. (c) MEG typing, 637 instances of 17 sentences, 19 participants; the trust layer equals always offer.

al. whether the participant confirmed each sentence before text-to-speech; describe that interaction if so.]

Error handling in BCIs. P300 and RSVP spellers commit a character when its posterior passes a threshold and stop early when confident [11, 12, 37], often with an LM prior [38–40]. Error-related potentials allow automatic error detection [13, 41–44]; asynchronous BCIs detect a "no-control" state [45–47]; myoelectric control rejects uncertain decisions [14]; adaptive decoders stay accurate over days by changing the decoder [48–52]. These rules are built into one decoder and paradigm; ours reads only the decoder's per-frame output, leaves the decoder fixed, and states a finite-sample bound.

Calibration, selective prediction and risk control. Temperature scaling fixes overconfidence [19] but degrades under shift [53]; the maximum softmax probability detects errors [54]; selective classification trades coverage for risk [20]. In BCI, confidence-aware abstention was shown for EEG speech commands [22], and calibrated alternatives improved brain-to-text fusion [23]; dialogue systems accept or reject utterances by cost-based thresholds [55]. Conformal risk control (CRC) bounds the expected value of a monotone loss [21, 56], Learn-then-Test (LTT) bounds the risk with high probability [26], weighted variants give approximate bounds under shift [57–59], and online methods give long-run guarantees for arbitrary sequences [27, 28, 60, 61]. Conformal methods were applied to speech recognition [24, 62] and to LM outputs and abstention [63–65], where LM stages can produce fluent unsupported text [66, 67].

Metrics and standards. Utility metrics [68], the practical bit rate [69] and text-entry metrics [70] charge for corrections, unlike information transfer rate [71]; LM prediction speeds up gaze typing [25, 72]. Our eWPM is close to the utility metric. NWB [73], BCI2000 and OpenViBE [74, 75], IEEE P2731 and P2794 [76–78] and decoder benchmarks [79, 80] each cover part of the stack; none defines a per-frame decoded-output format with confidence semantics.

3 Problem setting

Items, streams and actions. An *item* is one attempt to enter one sentence: a *trial* of the intracortical data, an *instance* (one participant typing one sentence) of the MEG data, or a composed *sentence* of the sEMG data. For item i, the decoder emits a stream of F_i frames of log-probabilities $\log p_t(v)$ over V tokens. An LM search returns $K = 10$ ranked candidate sentences with scores, from which we compute $\hat{p}_{\text{top1}}$, the normalised probability of the top candidate. The system then takes one action: COMMIT the top candidate (sent without further input); OFFER the top k candidates for the user to pick; OFFER_WORDS, the top candidate with uncertain words marked; or ABSTAIN, after which the user types with a fallback keyboard. Let $c_i \in \{0, 1\}$ indicate that the top candidate differs from the intended sentence (exact match after normalising case and punctuation), and $f_i \in \{0, 1\}$ the wrong-commit loss of the non-commit action ($f_i = 0$ for OFFER and ABSTAIN; for OFFER_WORDS, $f_i = 1$ if a wrong word is not marked).

The two error quantities. Over the n_{test} test items,

$$\text{wrong-commit rate} = \#\{\text{wrong commits}\}/n_{\text{test}} \approx \Pr(\text{commit} \wedge \text{wrong}), \tag{1}$$

$$\text{error among commits} = \#\{\text{wrong commits}\}/\#\{\text{commits}\} \approx \Pr(\text{wrong} \mid \text{commit}). \tag{2}$$

The first is the *marginal* risk: how often wrong sentences reach the recipient per sentence attempted. Our risk controllers bound it, because it is monotone in the commit threshold (Section 5.4). Committing rarely makes it small, so we always report it with the *commit rate*. The second is the *selective* risk that a user sees in committed text; our controllers do not bound it, and we report it next to every result. Table 4 (Appendix A) collects the symbols; $\alpha = 0.05$ is the target wrong-commit rate and $\delta = 0.1$ the failure probability of Learn-then-Test.

4 The NeuralTextStream format

NeuralTextStream (NTS) is the contract between a decoder and any consumer of its output (Figure 2). A stream of F frames over V tokens holds an $F \times V$ array `log_probs` of natural-log probabilities, the frame times, the ordered vocabulary, its type (`char`, `key`, `word`, `phoneme`, `command` or `other`) and the CTC blank index (or `null`). Optional fields give the frame period, latency, producer and a `meta` object with named key prefixes (`device.*`, `decoder.*`, `lm.*`, `calibration.*`, `adapter.*`).

Unmodified probabilities. A producer must emit the decoder's own output distribution and must not sharpen or smooth it, for example for display. Calibration is left to the consumer, which can fit it on data with known truth. A field is interpreted as a probability only if the stream declares it as one in `calibration.*`. Ground-truth text must not appear in the stream (a leakage rule), and raw signals and direct identifiers stay out of it (privacy rules). Appendix J summarises the serialisations, the conformance suite and the audit log.

Three device types. Table 1 lists the adapters (148, 412 and 568 lines of Python for MEG, intracortical and sEMG). Mapping the intracortical and sEMG decoders into NTS needed no change to the frame core of v0.1, a Development release written by the same authors; the trust harness added one optional field (the truth in the stream's token space) and provenance keys. Every other part of the pipeline is the same code for all devices, with parameters re-fitted per device (Table 7).

Table 1: Devices, data and streams. One adapter per device type maps decoder output into NeuralTextStream. All datasets are public. Test units are those of the trust evaluation.

	MEG (typing)	**Intracortical (speech)**	**Forearm sEMG (keys)**
Dataset (licence)	SpanishBCBL (CC BY-NC 4.0)	B2T '25 copy task (CC0)	Toronto sEMG (CC BY 4.0)
Participants	19 (healthy typists)	1 (T15, ALS)	19 (healthy typists)
Test unit	637 instances of 17 unique sentences	816 trials, 14 days	551 sentences composed from isolated presses
Decoder	own asynchronous ConvCTC	published online RNN; own phoneme-to-word decoder	own CNN
NTS vocabulary	`char` (CTC)	`phoneme` (CTC, 41 tokens)	`key` (W3C codes)
Frame	40 ms	80 ms	one per key press
Commit decision tested?	no (no right top-1)	yes	weakly (0–4% committed)

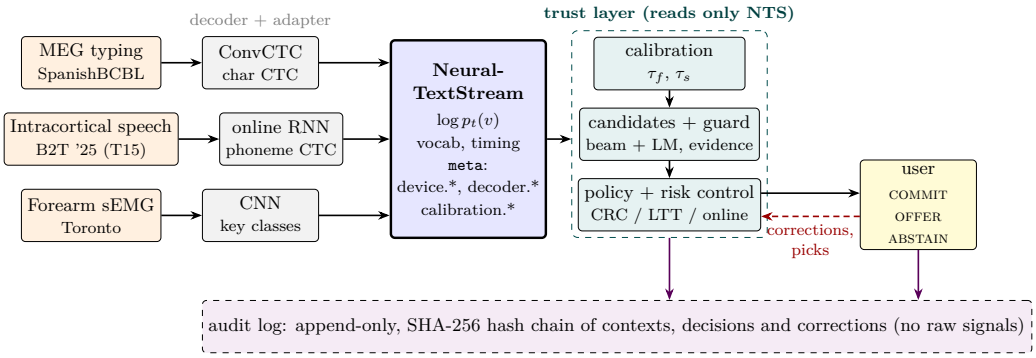


Figure 2: Architecture. Each device's decoder emits a NeuralTextStream through a small adapter. The trust layer reads only the stream: it calibrates confidence, builds candidate sentences with a neural-evidence guard, and chooses an action under a risk-control rule. The user's corrections and picks (dashed) feed the online re-calibration. Contexts, decisions and corrections can be written to an append-only, hash-chained audit log that stores no raw signals (Appendix J).

5 The trust layer

The trust layer reads one NeuralTextStream per item and returns one action. It never reads raw signals or decoder internals. Figure 6 in Appendix B shows the decision flow. All parameters below are fitted per device on that device's fit data; Table 7 lists the fitted values.

5.1 Calibration

Decoder log-probabilities are often over-confident. We fit a frame temperature τ_f that rescales every frame, $\tilde{p}_t(v) \propto p_t(v)^{1/\tau_f}$, by minimising the cross-entropy of the true token at the greedy CTC peak frames of the fit streams. We then fit a sentence temperature τ_s on the scores of the K-best list, $\hat{p}_{\text{top1}} = \text{softmax}(\text{scores}/\tau_s)_1$, by minimising the binary negative log-likelihood of "the top candidate is right" [19]. Both are golden-section searches over $\log \tau \in [\log 0.05, \log 20]$. We report the expected calibration error (ECE, 15 equal-width bins) of $\hat{p}_{\text{top1}}$. The frame temperature is applied *before* the LM search, so it changes the decoded text as well as the confidence (Section 7.4).

5.2 Candidates and the optional neural-evidence guard

Candidates come from a CTC prefix beam search with a character 6-gram LM trained on Tatoeba sentences, which returns a K-best list [35, 36]. For intracortical speech, a phoneme beam search is followed by a CMUdict lexicon, a Tatoeba word unigram and the same character LM. Because the LM can write fluent text from any input, a guard scores how much of the top candidate the LM invented: the maximum of the share of characters whose calibrated acoustic posterior is

below 0.5 (*low evidence*) and one minus the posterior-weighted word agreement between the top candidate and the top five (*K-best inconsistency*). A candidate whose guard risk exceeds a fitted limit cannot be committed. The guard was part of the configuration fixed before the evaluation runs; the ablation shows that it raised wrong commits on intracortical data and had no effect elsewhere (Section 7.4), and that its value lies in the signal-destroying controls (Section 7.5).

Gate on a published sentence. For the published intracortical decoder we have the stream and the published LM-rescored sentence, but not the decoder's LM or weights. We put the published sentence on top of the K-best list of our own phoneme-to-word decoder, with a score equal to our best score plus a fixed bonus of 2.0, and give each published word the acoustic confidence of the aligned word in our top candidate (0 if no word aligns). The guard and the policy then run unchanged. The gate thus needs the stream, the published text and a light reference decoder, but not the vendor's weights.

5.3 Decision policy

The policy chooses COMMIT, OFFER (with k candidates), OFFER_WORDS or ABSTAIN. Its thresholds (commit probability, maximum guard risk for a commit, offer size, abstain risk, fallback after an unused offer) are chosen on fit data from a grid to maximise eWPM (Section 6) subject to a wrong-commit rate of at most α on the fit data. This *plug-in* fit has no guarantee on new data, which motivates Section 5.4.

5.4 Finite-sample risk control for the commit threshold

The commit score s_i is the calibrated $\hat{p}_{\text{top1}}$ when the guard and the other policy rules allow a commit, and $-\infty$ otherwise. The layer commits when $s_i \geq \lambda$, and we use the loss

$$L_i(\lambda) = \max(c_i, f_i)\,\mathbf{1}[s_i \geq \lambda] + f_i\,\mathbf{1}[s_i < \lambda], \tag{3}$$

which upper-bounds the true loss and is non-increasing in λ, as CRC requires. The controlled risk is $R(\lambda) = \mathbb{E}[L(\lambda)]$; with $f \equiv 0$, it is the probability that an item is committed *and* wrong (Section 3). On n calibration items with known truth and empirical risk $\hat{R}_n$, CRC [21] chooses $\hat{\lambda}_{\text{CRC}} = \inf\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \leq \alpha\}$. If the calibration items and a new item are exchangeable, then $\mathbb{E}[L_{n+1}(\hat{\lambda}_{\text{CRC}})] \leq \alpha$, where the expectation is over the draw of the calibration items and the new item. The implementation searches the calibration scores and $+\infty$; this is valid and slightly conservative. We use $f \equiv 0$ (no OFFER_WORDS under CRC), so "never commit" is always feasible. For $n < 1/\alpha - 1$ (19 items for $\alpha = 0.05$) no finite threshold is feasible and the layer never commits. Learn-then-Test (LTT) [26] tests $H_0\colon R(\lambda) > \alpha$ along a fixed grid from 1.00 down to 0.00 in steps of 0.01 with exact binomial p-values and fixed-sequence testing. It gives $\Pr(R(\hat{\lambda}_{\text{LTT}}) \leq \alpha) \geq 1 - \delta$ over the calibration draw, where R is the population risk at the chosen threshold, not the rate on a finite test set. Because L_i is non-increasing in λ, any threshold above $\hat{\lambda}_{\text{CRC}}$ keeps the bound, so the deployable rule is $\max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}})$; "pure" CRC drops λ_{tuned}.

The separate-data requirement. Both guarantees hold only if the temperatures, the guard and the other policy thresholds, which define s_i, are fitted on data disjoint from the calibration items. Every CRC and LTT result in this paper meets this requirement: random splits draw calibration items from the test pool, and the deployment protocol (Section 6) splits the fit data into a part that fits the pipeline and a part that only calibrates. Our earlier analysis calibrated on the fit data itself; Section 7.7 reports what this changed. We use $\alpha = 0.05$ and $\delta = 0.1$ throughout.

5.5 Online re-calibration as a safeguard

Exchangeability can fail during use, for example when attention or electrode contact changes. The online controller re-tunes the commit threshold from user feedback; it is a safeguard whose guarantee does not need exchangeability. A corrected commit reveals a wrong commit, an offer reveals the truth when the user picks or retypes, and an abstention reveals nothing unless the final text is logged; the controller uses the *realised loss* and, when the final text is known, the counterfactual commit loss, possibly late. In our copy-task replays the truth comes from the cue sentence; error-related potentials are a possible implicit source [13]. We compare four methods (Table 11, Appendix K): `window` (CRC on the last 200 items with known truth) and `decay` (weighted CRC, weights 0.99^{age}) have no guarantee under shift; `aci` updates $\lambda_{t+1} = \max(0, \lambda_t + \eta(L_t - \alpha))$ with $\eta = 0.05$ [27, 28, 61], and `aci_window` applies the same kind of update to the level of a windowed CRC. For any sequence of T items, both bound the average observed loss by α plus a term of order $1/T$ that is bounded in advance only if scores are at most 1 and the non-commit loss is 0, and that grows with the feedback delay. These long-run bounds say nothing about short windows, and a wrong commit that the user does not notice (a *silent error*) is outside them. All online settings were fixed before the first evaluation run.

6 Evaluation protocol

6.1 Outcomes, user model and baselines

The primary safety outcome is the wrong-commit rate (Section 3), always reported with the commit rate and the error among commits. Speed is secondary. We simulate a user who reads the output, accepts or corrects a commit, picks from an offer, or types the sentence with a fallback keyboard. For item i with intended sentence y_i, the simulation returns the total time t_i (input, reading, selections, corrections, retyping) and the final text, and

$$\text{eWPM} = 60 \cdot \frac{\sum_i \mathbf{1}[\text{final text}_i = y_i] \ \text{words}(y_i)}{\sum_i t_i}. \tag{4}$$

Like the BCI utility metric [68], eWPM charges the time needed to fix errors. A *silent error* is a wrong final text that the user did not correct; it counts zero words, and we report its rate separately because eWPM does not price its harm.

The main user types with a gaze keyboard at 20 characters per minute and scans and selects one of $k = 3$ word alternatives in 4.5 s; both are matched to one eye-gaze user with ALS [25], and all other costs are assumptions (Table 5). The neural input runs at 159.3 (intracortical, measured), 75.8 (sEMG, median press period) and 152 (MEG) characters per minute. After a COMMIT, the user misses each wrong word with probability 0.05, which makes a silent error; after an OFFER, the main user sees every error and chooses the cheapest of picking, fixing or retyping. This asymmetry favours offers and is not measured. For the able-bodied MEG and sEMG typists, eWPM answers a hypothetical question about a gaze-typing user with a decoder of this error profile. Because only two costs are measured, we refit every policy under eight variants (fallback 10 or 40 characters per minute, selection costs $\times 0.5$ or $\times 2$, miss probability 0 or 0.2, and a *symmetric-offer* user who also misses wrong words in offers, with miss probability 0.05 or 0.2) and score the main-user policies under each variant without refitting (Appendix C).

Baselines: always commit the top candidate; always offer and never commit, the fair comparator for a commit rule because the layer can gain speed only by committing; a *plain threshold* on the calibrated $\hat{p}_{\text{top1}}$ without the guard; gaze typing alone (error-free, no reading time, which favours it); and gaze typing with *word completion*, where the keyboard shows the three most frequent words with the typed prefix (a unigram without the test sentences) and the user selects the intended word when this is faster (no glance cost, which favours it).

6.2 Datasets and splits

Intracortical speech (Brain-to-Text '25). The public copy-task file holds the online phoneme CTC logits, cue sentences and published LM-rescored sentences of participant T15, an implant recipient in the BrainGate2 clinical trial [2, 34]: 1,718 trials from 15 post-implant days. We use the 1,628 trials with the 125,000-word vocabulary (14 days); the 90 trials with the 50-word vocabulary (all of day 25 and 40 of day 27) are a different task and are excluded. The file has no time stamps, so the file order within each day is the trial order. We fit on the first half of each day (812 trials) and test on the second half (816); one sentence occurs in both. We evaluate our phoneme-to-word decoder (test WER 0.149) and the published sentence (WER 0.043), gated as in Section 5.

Forearm sEMG (Toronto). The dataset holds isolated key presses from 19 participants over two days [8, 81]. We compose held-out Tatoeba sentences from single presses with oracle timing and no inter-key context (570 fit, 551 test sentences). A press can recur across test sentences, so participant intervals are primary. A CNN classifies each press under three conditions: *session-calibrated* (day 1 plus half of the day-2 recordings, cross-fitted), *cross-session* (day 1 only) and *cross-subject* (leave-participants-out). Half A of the fit sentences fits the temperatures and the LM weight, half B the policy.

MEG typing (SpanishBCBL). Healthy participants typed memorised Spanish sentences in an MEG scanner [5, 82]. We split 19 participants' data by a hash of the sentence text into 101 training, 11 validation and 17 test sentences (637 test instances); the same people appear in all splits, and no test sentence is in training or LM text. All trust parameters are fitted on the 437 validation instances. Our asynchronous ConvCTC decoder (Appendix I) gets no key times at inference, but it uses key times as training labels and sentence bounds from the keyboard log. The trust harness keeps an insertion bonus of 1.0 tuned on raw streams; on frame-calibrated streams this adds spurious characters (top-1 CER 0.522 vs 0.475 with the validation-tuned bonus 0). With the validation-tuned beam, trust gave 3.14 instead of 3.20 eWPM.

Calibration data for CRC and LTT. Random splits of the test pool are exchangeable by construction and check the implementation. The *deployment protocol* calibrates on earlier data that played no other role. On Brain-to-Text '25, each day's fit trials are split into fit-A (temperatures, conformal sets, guard and policy; 411 trials) and a clean calibration set (401 trials), by alternating trials (split A) or by taking the earlier half as fit-A (split B). On sEMG, the policy grid uses the first 142 sentences of half B and the other 143 only calibrate; on MEG, 6 validation sentences (238 instances) fit the pipeline and 5 (199 instances) calibrate. As an exchangeable reference, we pool the calibration and test items and draw 1,000 random splits of the same sizes.

6.3 Controls and statistics

Decoder controls (shuffled MEG, zero input, permuted training labels) test that the decoders use the signal. *Signal-destroying* trust controls keep the stream format but remove the neural information (frame-shuffled and near-uniform streams, zero-input output, a permuted-label decoder); a stream-only layer should not commit these. The *intent-mismatch* control is the real stream of a different sentence by the same participant, which a stream-only layer cannot detect.

We report 95% percentile bootstrap CIs ($B = 2000$) and paired differences, resampled at the primary unit: days for Brain-to-Text '25 (14 clusters), participants for sEMG (19), and sentences and participants at once (two-way) for MEG. We name two primary endpoints per device, chosen in our first revision after the first evaluation runs: the wrong-commit rate of the trust layer and the paired eWPM difference to gaze typing. The always-offer, plain-threshold and

word-completion baselines, the user-model variants, the ablation and the deployment protocol were added after review, with settings fixed before they were run; their intervals are descriptive and not corrected for multiplicity. All decoder settings, user-model values, the policy grid, α, δ and the online step sizes were fixed before the evaluation runs. Every decoder and trust run uses one seed and one fit/test split.

7 Results

7.1 The decoders use the neural signal

Each decoder reads its input (Table 6 in Appendix D). Our asynchronous MEG decoder reached a test CER of 0.506 (greedy) and 0.475 (beam and LM tuned on validation); shuffled MEG gave 0.760 and zero input 0.925, and a text-only output that ignores the MEG reaches 0.688. In the trust harness the top-1 sentence was right for 2 of 637 instances. For context, the published asynchronous encoder reports 0.59 ± 0.02 with another split and without key-time training labels [6]. Our MEG results describe one weak decoder on 17 unique test sentences. **[TODO: seeds: test CER of MEG decoder seeds 1 and 2 (training in progress), seed SD next to seed 0 (0.506 / 0.475), and MEG trust numbers per seed.]** On Brain-to-Text '25, our phoneme-to-word decoder reached WER 0.149 with the right sentence first for 39% of test trials, and the published decoder WER 0.043 with 81%. On sEMG, balanced key accuracies (0.367, 0.240, 0.226) were well above uniform chance (0.037; permuted labels 0.032), and after the LM the right sentence was first for 46%, 12% and 4% of composed sentences.

7.2 The layer cuts wrong commits; its speed gain over always offering is small

Table 2 and Figure 1 give the main result.

Intracortical speech. On our decoder, the trust layer cut the wrong-commit rate from 0.610 (always commit) to 0.059 [0.045, 0.071]. It committed 23% of trials, and 48 of these 189 commits were wrong (error among commits 0.25 [0.19, 0.32]). It was 0.18 eWPM (1.6%) faster than always offering [+0.12, +0.24]; the difference to always committing was [−0.16, +0.33]. With CRC on the clean calibration set (split A), the rate fell to 0.0245 at a commit rate of 0.10, still faster than always offering [+0.07, +0.16], with error among commits 0.24. On the published sentence, whose WER is only 0.043, always committing gave a wrong-commit rate of 0.194, because one wrong word makes a sentence wrong. The gate reduced it to 0.044 [0.027, 0.067] (36 of 386 commits wrong, 0.09), at a cost of 4.3% against always committing [−1.01, −0.67] and 0.66 eWPM (3.6%) faster than always offering [+0.53, +0.80]. All policies were far faster than gaze typing, with or without word completion.

Forearm sEMG (composed sentences). The layer cut the wrong-commit rate from 0.541, 0.882 and 0.962 to 0.004, 0.004 and 0, committing 21, 10 and 0 of 551 sentences, so it was nearly always offer (paired differences [+0.01, +0.03], [−0.00, +0.01], 0). Only the session-calibrated decoder clearly beat gaze typing [+2.73, +4.84] and gaze typing with word completion [+0.74, +2.99]. For the cross-session and cross-subject decoders, the difference to gaze typing was [−0.05, +0.98] and [−0.45, +0.10], and word completion was faster than the layer ([−1.99, −0.95], [−2.35, −1.84]). CRC on clean calibration data raised the commit rate to 0.24 and 0.09 for at most +0.05 eWPM; on cross-session, 61% of these commits were wrong.

MEG (negative control for the commit decision). The intended sentence was in the 10-best list for 1.6% of instances, and always committing gave a wrong-commit rate of 0.997. The trust layer committed nothing, so it equals always offering. Its difference to gaze typing was [−0.11, +0.26] two-way (sentences [−0.05, +0.15], participants [−0.08, +0.22]): no detectable

Table 2: Safety and speed (test sets). 95% cluster-bootstrap CIs over days (B2T '25), participants (sEMG) or sentences × participants (MEG). Wrong-commit and commit: per item; Err. | commit: wrong commits / commits; Silent: uncorrected wrong final texts per item. Trust: configuration fixed before evaluation (with guard). Plain threshold: calibrated $\hat{p}_{\text{top1}}$, no guard (= trust on sEMG, MEG). CRC: fit-A pipeline, clean calibration (B2T split A, $n = 401$; sEMG 143; MEG 199). "—": no commits.

Device	Policy	eWPM [CI]	Wrong-commit [CI]	Commit	Err. | commit	Silent
B2T '25, our decoder	always commit	11.45 [10.69, 12.32]	0.610 [0.569, 0.648]	1.00	0.61	0.044
	always offer	11.37 [10.66, 12.20]	0	0	—	0
	plain threshold	11.51 [10.77, 12.38]	0.036 [0.025, 0.047]	0.17	0.22	0
	trust	11.55 [10.82, 12.38]	0.059 [0.045, 0.071]	0.23	0.25 (48/189)	0.002
	trust, CRC (clean cal.)	11.48 [10.78, 12.28]	0.025 [0.015, 0.035]	0.10	0.24	0.001
B2T '25, published	always commit	19.61 [17.82, 21.35]	0.194 [0.145, 0.253]	1.00	0.19	0.025
	always offer	18.11 [16.61, 19.61]	0	0	—	0
	plain threshold	18.62 [17.03, 20.22]	0.025 [0.014, 0.039]	0.38	0.06	0.001
	trust gate	18.77 [17.15, 20.40]	0.044 [0.027, 0.067]	0.47	0.09 (36/386)	0.004
	trust, CRC (clean cal.)	18.79 [17.16, 20.44]	0.048 [0.029, 0.070]	0.49	0.10	0.005
B2T '25	gaze / gaze + completion	4.08 [4.05, 4.12] / 5.84 [5.77, 5.90]	0	—	—	0
sEMG session-cal.	always commit	7.81 [6.68, 9.00]	0.541 [0.468, 0.623]	1.00	0.54	0.060
	always offer	8.01 [6.96, 9.11]	0	0	—	0
	trust	8.02 [6.97, 9.13]	0.004 [0, 0.009]	0.04	0.10 (2/21)	0
	trust, CRC (clean cal.)	8.07 [7.03, 9.18]	0.040 [0.024, 0.055]	0.24	0.17	0.005
sEMG cross-session	always commit	4.10 [3.59, 4.73]	0.882 [0.833, 0.927]	1.00	0.88	0.149
	always offer	4.67 [4.22, 5.25]	0	0	—	0
	trust	4.68 [4.22, 5.25]	0.004 [0, 0.009]	0.02	0.20 (2/10)	0
	trust, CRC (clean cal.)	4.66 [4.20, 5.25]	0.056 [0.040, 0.073]	0.09	0.61	0.007
sEMG cross-subject	always commit	3.53 [3.19, 3.92]	0.962 [0.940, 0.980]	1.00	0.96	0.167
	trust (= always offer, = CRC)	4.08 [3.82, 4.38]	0	0	—	0
	trust, LM weight re-tuned	4.84 [4.25, 5.51]	0.002 [0, 0.006]	0.01	0.17 (1/6)	0
sEMG	gaze / gaze + completion	4.27 [4.22, 4.32] / 6.19 [6.11, 6.26]	0	—	—	0
MEG (asynchronous)	always commit	2.58 [2.31, 2.88]	0.997 [0.986, 1.000]	1.00	1.00	0.235
	trust (= always offer)	3.20 [3.02, 3.43]	0	0	—	0
	trust, CRC (clean cal.)	3.20 [3.01, 3.42]	0.028 [0.008, 0.061]	0.03	0.95	0.005
MEG	gaze / gaze + completion	3.15 [3.02, 3.30] / 4.10 [3.89, 4.33]	0	—	—	0

difference. Gaze typing with word completion was faster [−1.14, −0.64]. CRC on clean calibration data committed 3.0% of instances, and 95% of these commits were wrong.

7.3 Speed rankings depend on the user model; wrong-commit rates do not

Figure 3 refits every policy under eight user-model variants. First, the trust layer was at least as fast as always offering in every variant and condition; on Brain-to-Text '25 all intervals of the difference were above 0. Second, the comparison with always committing depends on the user model: on our intracortical decoder, always committing was at least as fast in 5 of 8 variants (fallback 40 characters per minute, selection ×2, miss probability 0, both symmetric-offer variants), and on the published decoder it was the fastest policy in all 9, by 0.46–1.47 eWPM. On sEMG and MEG, the gain over always committing came from offers and disappeared with miss probability 0 or symmetric offers. Third, gaze typing with word completion was faster than the layer in every variant on MEG and in all but one on the two weaker sEMG decoders (with a 1 s glance cost per look: 3.16 and 4.84 eWPM). On MEG, the comparison with plain gaze typing flipped either way across variants (40 characters-per-minute fallback [−1.36, −0.89]; 10 characters per minute [+0.21, +0.45]). In contrast, the wrong-commit rate of the refitted trust layer was unchanged in 49 of 54 condition–variant pairs and at most 0.059 in all. Silent errors rose under symmetric offers (0.02–0.17), because offers then also let errors through. Policies fitted under the main user lost at most 0.2 eWPM against refitting, except with symmetric offers and miss probability 0.2 on the weak decoders (MEG 1.18 vs 2.75).

7.4 What makes the layer safe: sentence calibration and a threshold, not the guard

Calibration. Sentence ECE fell from 0.448 to 0.067 on our intracortical decoder (frame temperature 3.78; the ECE of per-frame phoneme peaks rose from 0.090 to 0.130) and from 0.154, 0.328 and 0.500 to 0.148, 0.072 and 0.083 on sEMG (Figure 4a). The calibrated top-1 probabil-

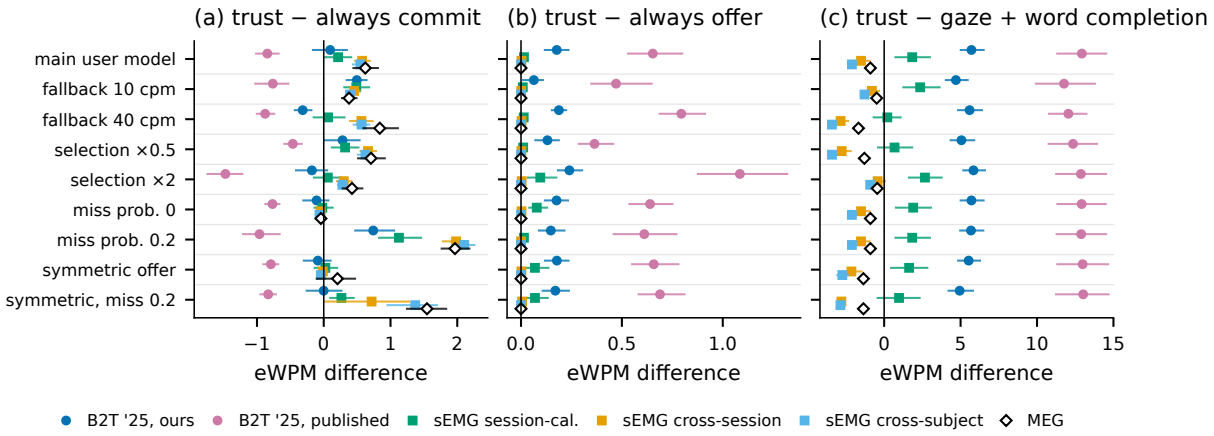


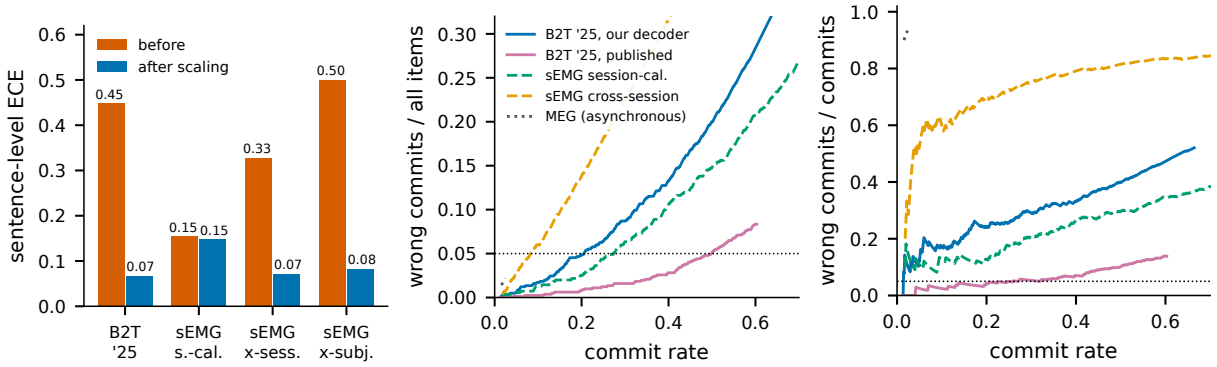
Figure 3: User-model sensitivity. Paired eWPM difference between the trust layer and (a) always committing, (b) always offering and (c) gaze typing with word completion, all policies refitted under each variant (rows); 95% cluster-bootstrap CIs at the primary unit.

ity predicted a wrong top sentence with AUROC 0.797 (intracortical) and 0.83, 0.76 and 0.67 (sEMG); we did not compute CIs for these AUROCs or reliability diagrams.

Ablation (Table 8, Appendix F; nine variants, six conditions). (1) The *sentence* temperature keeps wrong commits low: without it, the rate rose on 4 of 6 conditions (our intracortical decoder 0.076; sEMG 0.040, 0.018, 0.044). (2) The *frame* temperature changes decoding, because it is applied before an LM search whose weight was tuned on raw streams. On sEMG cross-subject, decoding the raw stream gave top-1 CER 0.369 instead of 0.479 and 4.75–4.76 instead of 4.08 eWPM; re-tuning the LM weight on the calibrated streams gave 4.84 [4.25, 5.51] at CER 0.360. This is above gaze typing as a point value, but the interval of the difference [−0.02, +1.25] includes 0, and it stays below gaze typing with word completion; our earlier conclusion that this decoder stays below gaze typing was an artefact of the LM weight. (3) The guard did not help on real data: without it, the wrong-commit rate fell from 0.059 to 0.036 on our intracortical decoder (paired [−0.036, −0.011]; eWPM [−0.10, +0.04]) and from 0.044 to 0.025 on the published gate ([−0.034, −0.006]; eWPM [−0.20, −0.10]), and changed nothing on sEMG and MEG. Its AUROC for a wrong top sentence (0.702) was below that of $\hat{p}_{\text{top1}}$. Conformal sets and OFFER_WORDS changed little.

Why the guard stays in the reported configuration. We report the configuration fixed before evaluation, with the plain threshold next to it (Table 2): choosing it from a test-set ablation would bias the reported rates downwards, and the plain threshold was not run on the control streams, where the guard separated signal-destroying streams from real ones (Section 7.5). A future pre-specified configuration should use the plain threshold, with the guard as an optional check that must pass the controls.

The commit trade-off. As the threshold varies (Figures 4b and 4c), the published decoder can commit about half of its sentences at the 0.05 marginal level and our decoder about a fifth. The error among commits of our decoder falls to 0.1–0.2 only below about 10% commits and rises to about 0.5 at 60%, while that of the published decoder stays near or below 0.15; on sEMG cross-session, most commits are wrong at all but the lowest rates. A rule that controls the marginal risk can therefore accept a high error among commits on a weak decoder.



(a) Sentence-level ECE. **(b)** Marginal risk. **(c)** Selective risk.

Figure 4: Calibration and the commit trade-off. (a) Sentence-level expected calibration error before and after temperature scaling (test sets). (b) For each commit threshold on the calibrated score: share of test items committed (x) and committed wrongly (y). (c) The same thresholds: error among commits. Items blocked by the guard or other policy rules are never committed. Curves start at 10 commits. Dotted line: $\alpha = 0.05$. sEMG cross-subject is absent because no candidate passed the guard.

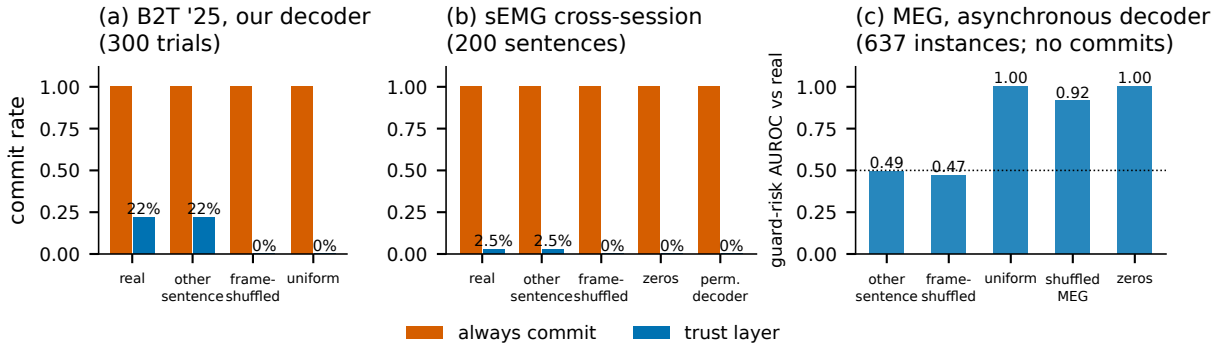


Figure 5: Controls. (a, b) Commit rate of always commit (orange) and the trust layer (blue) on real input, the intent-mismatch control (other sentence) and signal-destroying controls: (a) B2T '25, our decoder, 300 test trials; (b) sEMG cross-session, 200 test sentences. On controls other than "other sentence", any commit is wrong. (c) MEG, 637 instances: AUROC of the guard risk for separating each control from real input; the MEG layer committed nothing.

7.5 Controls: signal-destroying streams are not committed; intent mismatch is not detectable

Always commit wrote a sentence for every control stream (on MEG, 85–100% fluent Spanish). The trust layer committed none of the signal-destroying controls on Brain-to-Text '25 (300 trials) or sEMG (200 sentences per condition) (Figure 5a, b), an informative result because the same layer committed 22% and 2.5–4% of the real inputs in the same audit. On Brain-to-Text '25, the guard risk separated frame-shuffled and uniform streams from real ones with AUROC 0.981 and 0.998. On MEG, the layer committed nothing on real input either, so its zero commits on controls carry no information; the guard risk separated shuffled MEG, zero input and uniform streams (AUROC 0.92, 1.00, 1.00), but not frame-shuffled streams (0.47) (Figure 5c). Given the real stream of another sentence, the layer committed at the matched rate on our intracortical decoder (22%) and on sEMG: nothing in the stream reveals the intent. The published gate committed none of these, but only because its text came from the matched trial.

7.6 CRC bounds the mean marginal risk, not the error among commits

On random splits of the test pool, which are exchangeable by construction, CRC held the mean wrong-commit rate at or below α at every calibration size (Table 12 in Appendix M); this checks the implementation, not the assumption. With 408 calibration trials, single splits exceeded 0.05 in 43% (our decoder) and 46% (published) of cases, partly from test-sampling noise (standard error about 0.011); LTT had a 90th percentile of 0.052 and 0.054, against 0.069 and 0.071 for CRC. Because the pool is the population of this experiment, we can compute the risk that LTT bounds, the rate of the chosen threshold on the whole pool: over 200 splits it exceeded α in 0.5% and 1.5% of LTT splits (within $\delta = 0.1$) and in 33% and 29% of CRC splits, which CRC does not promise to avoid. Small calibration sets keep the mean valid but make the layer commit rarely: with 20 trials, CRC had a 90th percentile of 0.089 and never committed in 17% of splits, and LTT cannot commit below 45 items ($0.95^n > \delta$). A session-start block of the first 20 trials of each day gave CRC rates of 0.009 (ours) and 0.025 (published) at commit rates of 0.04 and 0.32, and 6 of 14 days never committed; on sEMG, 9 of 18 participants never committed. The error among commits did not change with the calibration size: about 0.25 (ours), 0.10 (published) and 0.93 on MEG, where CRC still committed 2.1% of instances because this is within the budget. This is control without utility; the deployable rule $\max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}})$ inherits the plug-in threshold, which committed nothing on MEG.

7.7 No detectable shift after clean calibration: the earlier drift was a data-reuse artefact

In our earlier analysis, CRC calibrated on the fit half of each Brain-to-Text '25 day gave 0.0625 on the test half, and pure CRC on sEMG cross-session gave 0.0635; we read this as within-day drift. But those calibration items had also fitted the temperatures, the conformal sets and the policy grid, which violates the separate-data requirement (Section 5.4). With clean calibration data (Table 3), offline CRC gave 0.0245 and 0.0600 on our decoder (splits A and B), 0.0478 and 0.0490 on the published gate, and 0.0563 and 0.0399 on sEMG cross-session and session-calibrated; on sEMG cross-subject it could not commit. No value above α was significantly above the rate that exchangeable splits of the same items give (one-sided $p \geq 0.21$); in split A the test rate was above the exchangeable reference ($p = 0.012$) but below α. The earlier 0.0625 thus came from data reuse and split-to-split variation, and we do not claim within-day drift for these data.

The online methods kept the long-run rate at 0.0245–0.0527 (Brain-to-Text '25) and 0.0381–0.0526 (sEMG), at eWPM within 0.06 of offline CRC. The `aci` and `aci_window` certificates were 0.038–0.053, within loose a-priori bounds of 0.059–0.081. Shuffling the test order changed the online rates by at most 0.007, and a feedback delay of 10 items by at most 0.009. When only noticed errors were fed back (delay 10), `aci` reached 0.0527 (Brain-to-Text '25, split B) and 0.0581 (sEMG cross-session) while its certificate on the observed losses was 0.0509 and 0.0499: silent errors look correct to the controller. On these data the online controller is a safeguard with an any-sequence guarantee, not a fix for an observed failure.

8 Discussion

The layer is a safety mechanism, not a speed-up. Against always committing, the trust layer cut wrong commits by an order of magnitude on intracortical speech. Against always offering, the best policy that never commits, it was 1.6% and 3.6% faster, for 59 and 44 wrong commits per 1,000 sentences. Whether it beat always committing depended on unmeasured user costs, and on weaker decoders a gaze keyboard with simple word completion was faster still, so we make no general speed claim. What holds across devices and user models is the safety result: the layer commits only when calibrated evidence supports it, its wrong-commit rate is a stated

Table 3: Wrong-commit rate on later data with clean calibration. Reused: our earlier offline CRC, calibrated on data that also fitted the pipeline. Clean CRC: calibration data disjoint from all fitted parts (95% CI over days or participants). Exch. ref.: mean [5th, 95th percentile] over 1,000 random splits of the pooled calibration and test items; p: share of these splits at least as high as observed. Online: warm start on the clean calibration data, then the test data in order, delay 0. Cert. / a priori: `aci` bound after the fact and in advance. sEMG cross-subject: no method committed.

		Offline				Online				
Data (n_{test})	**Reused**	**Clean CRC [CI]**	**Exch. ref.**	p	**LTT**	`window`	`decay`	`aci`	`aci_w.`	**Cert. / a priori**
B2T '25 ours, A (816)	0.0625	0.0245 [0.014, 0.035]	0.018 [0.014, 0.022]	0.012	0.0270	0.0245	0.0257	0.0270	0.0245	0.043 / 0.068
B2T '25 ours, B (816)	0.0625	0.0600 [0.049, 0.071]	0.049 [0.029, 0.071]	0.21	0.0515	0.0478	0.0453	0.0527	0.0490	0.053 / 0.066
B2T '25 published, A (816)	0.0490	0.0478 [0.029, 0.071]	0.049 [0.029, 0.074]	0.54	0.0294	0.0429	0.0392	0.0478	0.0453	0.048 / 0.059
B2T '25 published, B (816)	0.0490	0.0490 [0.030, 0.074]	0.050 [0.031, 0.074]	0.50	0.0306	0.0441	0.0392	0.0478	0.0453	0.048 / 0.059
sEMG cross-session (551)	0.0635	0.0563 [0.040, 0.074]	0.045 [0.018, 0.080]	0.30	0.0309	0.0472	0.0381	0.0526	0.0490	0.053 / 0.081
sEMG session-cal. (551)	0.0417	0.0399 [0.024, 0.056]	0.047 [0.018, 0.082]	0.60	0.0218	0.0490	0.0454	0.0490	0.0490	0.049 / 0.072

quantity with a bound, and it costs little or no speed relative to always offering. The user or clinician chooses α, and the layer reports the commit rate and error among commits that follow. On sEMG and MEG it reduced to always offering, the correct behaviour for decoders whose top sentence was rarely right; it cannot create information that the decoder lacks.

Marginal vs selective risk. The controlled quantity bounds how often wrong text reaches the recipient per sentence attempted, not how often committed text is wrong (one commit in four on our intracortical decoder; 95% of the few CRC commits on MEG). We chose it because it is monotone in the threshold, which CRC needs, and because it is the right budget for harm per attempt. A selective-risk controller, for example LTT conditional on the number of commits with a minimum commit rate, is a natural complement that we did not evaluate. Each guarantee should be stated with its assumptions, its target and the measured silent-error rate (Appendix L).

Data reuse can fake a drift failure. Our earlier analysis read offline CRC above target on later data as within-day drift; with calibration data that played no other role, the excess disappeared into split-to-split variation. CRC needs calibration items exchangeable with new items *for a fixed score function*; when the same items also fit the temperatures and the policy, their scores look better calibrated than new scores, and the threshold is too permissive. The mistake is easy in BCI pipelines, where data are scarce; papers should state which data fitted each stage and compare any claimed shift with an exchangeable reference.

Strong decoders still need a gate. The published Brain-to-Text '25 decoder has a WER of 0.043, yet 19% of its sentences were wrong. Gating its text by agreement with the neural evidence kept 96% of its speed and reduced wrong commits to 4.4% (2.5% with a plain threshold), using only the stream, the published text and a light reference decoder. This is what a vendor-neutral stream format enables: one gate and controller for three device types, with adapters of 148 to 568 lines.

Calibration and adaptation. A frame temperature applied before an LM search requires re-tuning the LM weight. On sEMG, in-session decoder data gave 71% higher eWPM than another day's data (8.02 vs 4.68): the layer made both safe, but only adaptation made them fast. A 20-item session-start block is too small for a fixed threshold, so it should seed an online controller.

9 Limitations

1. **One implant participant** in a copy task, with online logits we cannot re-run, is the only non-trivial test of the commit decision; our intervals cover days, not people.

2. **Composed sEMG sentences**: isolated presses with oracle timing, recurring presses, and test sentences from the LM's corpus (295 of 551 have a near variant in the LM text; removing them changed trust eWPM by at most −0.04, Appendix G). **MEG is a negative control**: 17 unique test sentences, and a decoder that uses key times as training labels and keyboard-log sentence bounds.
3. **No live users.** Only two user costs are measured, in one gaze-typing user; the miss probability is unmeasured; the MEG and sEMG populations do not match; speed rankings change across plausible variants; the word-completion baseline uses only a unigram. A user study needs ethics approval.
4. **One seed and one split per run.** MEG decoder seeds 1 and 2 are still training; sEMG seeds, repeated Brain-to-Text '25 fit/test assignments and a k-fold over MEG sentences were not run. The two deployment splits moved offline CRC from 0.0245 to 0.0600. Trial order is file order (no time stamps), so "no detectable within-day shift" holds for this order only.
5. **Scope of the guarantees.** They are marginal or long-run (43–46% of single CRC splits exceeded 0.05), count only observed errors, and do not bound the error among commits. Stream-only gating cannot detect intent mismatch; the loss is exact match (we did not test a tolerant loss); the audit log is tamper-evident, not tamper-proof.

10 Conclusion

A trust layer that reads only a per-frame output stream can decide whether a neural text-entry decoder commits, offers or abstains under a stated bound on the wrong-commit rate. On one intracortical participant it turned frequent wrong commits into rare ones at little or no speed cost relative to always offering, while one committed sentence in four (our decoder) or eleven (published decoder) was still wrong; on composed sEMG sentences and a weak MEG decoder it almost never or never committed. With calibration data that played no other role, conformal calibration met its target on later data up to sampling variation; an apparent drift failure was data reuse. Next steps are a gaze-typing user study, a controller for the error among commits, and the public benchmark that the release starts.

11 Ethics, data and code availability

This study uses only public, de-identified datasets; no new data were collected and no experiment with human participants was run, so no new ethics approval was required. The intracortical data come from a participant in the BrainGate2 pilot clinical trial [2]; we use only the released copy-task logits and sentences and did not attempt re-identification. **[TODO: author decision: final wording and the venue's policy on secondary use of public clinical-trial data.]** No raw recordings or participant data are redistributed; Table 9 lists all datasets and licences. The NeuralTextStream specification, conformance suite, reference implementation, benchmark scorer and all experiment scripts are at `https://github.com/zhangzibo/fidetype` (Apache-2.0), without weights trained on SpanishBCBL or copied Brain2Qwerty code; every table and figure is generated from committed result files. **[TODO: tag the paper commit and archive it (e.g. Zenodo DOI).]** Decoders were trained on one AMD RX 7800 XT GPU; trust-layer runs use a CPU. **[TODO: report trust-layer run time per decision.]**

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[TODO: funding.] AI coding and writing assistants were used under human review; all numbers come from committed scripts and result files. **[TODO: author: final AI-assistance wording.]**

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A Notation

Table 4: Notation.

Symbol	Meaning	Symbol	Meaning
F_i	frames in the stream of item i	λ	commit threshold on s_i
V	vocabulary size of the stream	s_i	commit score of item i (Section 5.4)
K	size of the candidate list ($K = 10$)	n	number of calibration items
k	number of offered candidates	T	number of items in an online sequence
τ_f, τ_s	frame and sentence temperatures	η, γ	step sizes of the online controllers
c_i, f_i	commit loss, non-commit loss	α, δ	target risk, LTT failure probability

B Decision flow

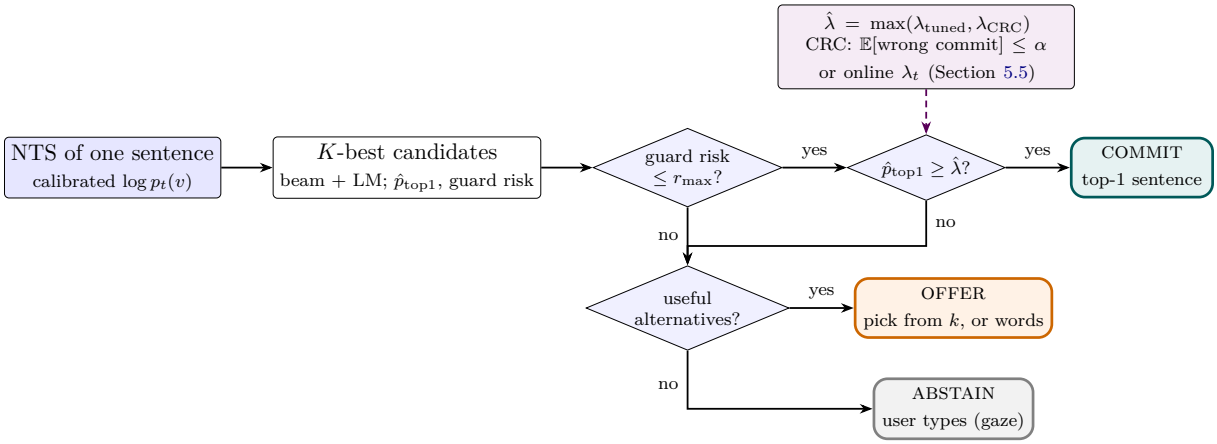


Figure 6: Trust decision for one item. Candidates come from the calibrated stream. A candidate is committed only if the neural-evidence guard accepts it and its calibrated confidence reaches the threshold $\hat{\lambda}$, which is the larger of the eWPM-tuned threshold and the CRC threshold (or the online threshold λ_t). Otherwise the layer offers alternatives or abstains, and the user types with the fallback keyboard.

C User model, variants and baselines

Table 5 lists the parameters of the main simulated gaze-typing user (word model: the user fixes single words). Only the fallback typing speed and the per-slot scan-and-select time are matched to measurements of one eye-gaze user with ALS [25]; all other values are assumptions.

Variants. Each variant changes one group of parameters and refits every policy: fallback speed 10 or 40 characters per minute (main 20); all selection, scanning, word-selection and navigation times scaled by 0.5 or 2; miss probability of a wrong committed word 0 or 0.2 (main 0.05). In the *symmetric-offer* variants, an offer without the intended sentence makes the user fix the closest candidate word by word, missing each wrong word with the same probability as after a commit; there is no oracle choice between fixing, repeating and retyping, and full retyping is disabled for all actions. In the *misspecified* evaluation, the trust and always-offer policies fitted under the main user are scored under each variant without refitting.

Gaze typing with word completion. The user types letter by letter at the fallback speed. Before each letter, the keyboard shows the three most frequent vocabulary words (counts ≥ 2) that start with the typed prefix; there is no sentence context. If the intended word is shown and selecting it is faster than typing the rest of the word and its space, the user selects it at

the scan-and-select cost for three slots plus the word-selection time (4.5 s at the main costs, close to the 4.7 s measured for one word choice [25]). The main baseline charges no time for looking at the list, which favours it; the *glance* variant charges 1 s per look. Unigram corpora: Brain-to-Text '25, the word unigram of the decoder (Tatoeba English); sEMG, Tatoeba English without the held-out test pool; MEG, Tatoeba Spanish without the SpanishBCBL sentences.

Table 5: Main simulated-user parameters.

Parameter	Value	Basis
Fallback (gaze keyboard) speed	20 characters/min	measured, eye-gaze user LP1 [25]
Scan and select one of k word alternatives	$1.0 + 1.0 \log_2(k+1) + 1.5$ s (4.5 s for $k = 3$)	sum matched to 4.7 s [25]
Select one of k offered sentences	$1.0 + 0.5 \log_2(k+1)$ s	assumption
Reading time	0.3 s per word, 0.5 s per candidate	assumption
Navigate to and delete a wrong word	3.0 s (2 selections)	assumption
Miss probability of a wrong, unmarked word (commit, OFFER_WORDS)	0.05	assumption (no measurement)
Errors in offered candidates	always seen (main model)	assumption
Full retype overhead	1.5 s; chosen when cheaper than fixing words	assumption (rational user)
Abstain overhead	0.5 s, then fallback typing	assumption
Neural input speed, MEG / B2T '25 / sEMG	152 / 159.3 / 75.8 characters/min	device setting

D Decoder quality

Table 6 gives the decoder metrics and input controls. Excluding the 6 MEG test sentences with a close training variant (CER ≤ 0.25 to a training sentence) gave 0.519 (greedy) and 0.492 (beam), so memorisation of near variants explains at most 0.02 of the CER. On sEMG, plain key accuracies (0.299, 0.168, 0.209) are below the majority-class rate (space, 0.338) because the CNNs rarely predict space; balanced accuracy is the fair metric.

Table 6: Decoder quality and input controls (test sets). Exact: share of items whose top-1 sentence is right; in K-best: share whose intended sentence is in the 10-best list. MEG: the trust harness keeps an insertion bonus of 1.0 tuned on raw streams; the validation-tuned bonus of 0 gives the 0.475 row (Section 6). sEMG chance: majority class 0.338, uniform 0.037. "—": not applicable or not computed.

Device / decoder	CER	WER	Exact / in K-best	Controls and floors
MEG, asynchronous, greedy	0.506	0.842	—	shuffled 0.760, zeros 0.925
MEG, beam + LM tuned on validation	0.475	0.809	0.005 / —	text-only floor 0.688
MEG, beam in the trust harness	0.522	—	0.003 / 0.016	—
B2T '25, our phoneme-to-word decoder	0.095	0.149	0.390 / 0.623	—
B2T '25, published decoder	0.031	0.043	0.806 / —	—
sEMG keys: accuracy (balanced)	0.299 (0.367) / 0.168 (0.240) / 0.209 (0.226)		—	permuted labels 0.022 (0.032)
sEMG sentences after LM	0.123 / 0.398 / 0.479	0.214 / 0.579 / 0.681	0.459 / 0.118 / 0.038	in K-best 0.64 / 0.21 / 0.08

E Per-device settings

Table 7 lists the parameters that the shared implementation fitted or used per device. The policy grid and all decoder settings are in the experiment configurations in the repository.

Table 7: Per-device settings of the shared trust-layer implementation. Fitted on each device's fit data (MEG: validation set). $\tau_s = 20.0$ is the upper end of the search interval. Max. guard risk: largest guard risk that still allows a commit.

	MEG	B2T '25, ours	B2T '25, published gate	sEMG (cal. / x-sess. / x-subj.)
Frame temperature τ_f	3.30	3.78	3.78	1.25 / 1.78 / 2.17
Sentence temperature τ_s	20.0	4.56	1.72	2.42 / 5.42 / 20.0
LM weight (beam)	0.5 (insertion bonus 1.0)	1.0 (word LM; edit penalty 12, word bonus 3)	as ours	1.0 / 1.0 / 1.5
Candidate list K	10	10	published sentence + our 10 (bonus +2.0)	10
Plug-in commit threshold	0.5	0.5	0.7	0.7 / 0.3 / 0.3
Max. guard risk for commit	0.3	0.5	0.5	1.0 / 1.0 / 0.1
Offer size k	1	2	1	2 / 2 / 1
Fallback after offer	offer	offer	offer	offer
Guard	low-evidence threshold 0.5; consistency over the top 5; risk = max(low evidence, 1 − consistency), weights 1			
Adapter (lines of Python)	148	412	412	568

F Ablation

Table 8 refits each variant on the condition's fit data and scores it on the same test items. *No calibration*: no temperature. *Confidence only*: both temperatures fitted, but the beam decodes the raw stream. *No sentence temperature*: the frame temperature is applied before the beam and the sentence calibrator is the identity. *No conformal sets*: OFFER shows the fitted top k. The OFFER_WORDS rows use a smaller common grid so that they are comparable. *LM weight re-tuned*: on sEMG, the LM weight is chosen on fit-A CER with the frame-calibrated streams (cross-session 1.0 to 0.5, cross-subject 1.5 to 0.5); on MEG, the validation-tuned beam (LM weight 0.25, insertion bonus 0). Top-1 CER on sEMG cross-subject: 0.479 (trust), 0.369 (raw decoding), 0.360 (LM weight re-tuned).

Table 8: Unified ablation (eWPM / wrong-commit rate; point values; "=": equal to trust). CIs and paired differences at the primary unit are in the repository (`ablation/results/tables.md`).

Variant	B2T '25 ours	B2T '25 published	sEMG s.-cal.	sEMG x-sess.	sEMG x-subj.	MEG
trust (as reported)	11.55 / 0.059	18.77 / 0.044	8.02 / 0.004	4.68 / 0.004	4.08 / 0	3.20 / 0
no calibration	11.57 / 0.071	18.73 / 0.043	8.15 / 0.045	4.75 / 0.013	4.76 / 0.005	3.04 / 0.005
confidence only (raw decoding)	11.54 / 0.074	18.74 / 0.047	8.07 / 0.018	4.76 / 0.007	4.75 / 0	3.04 / 0
no sentence temperature	11.50 / 0.076	18.66 / 0.032	8.11 / 0.040	4.67 / 0.018	4.07 / 0.044	3.20 / 0
no guard	11.51 / 0.036	18.62 / 0.025	=	=	=	=
no conformal sets	11.53 / 0.059	=	8.07 / 0.020	=	=	=
OFFER_WORDS off (small grid)	11.25 / 0.032	18.62 / 0.025	7.98 / 0.004	4.64 / 0	4.08 / 0	3.20 / 0
OFFER_WORDS on (small grid)	11.32 / 0.054	18.59 / 0.026	7.99 / 0.015	4.64 / 0.005	4.08 / 0	3.20 / 0
LM weight re-tuned	—	—	8.02 / 0.004	4.73 / 0.013	4.84 / 0.002	3.14 / 0
CRC, clean cal. (wrong-commit, off → on)	0.025 → 0.025	0.048 → 0.048	0.004 → 0.040	0.004 → 0.056	0 → 0	0 → 0.028
gaze typing only	4.08	4.08	4.27	4.27	4.27	3.15

G LM leakage check

A near variant is a sentence within a character edit distance of 0.25 times its length, after normalisation, in the 600,000-sentence training slice of the English character LM. Of the 1,717 unique Brain-to-Text '25 cue sentences, 22 have an exact copy and 310 an exact or near variant; of the 551 sEMG test sentences, none has an exact copy and 295 have a near variant (Tatoeba is heavily templated). We rebuilt the LMs without these sentences (849 removed for Brain-to-Text '25, plus 910 from the word unigram; 2,895 for sEMG, including near variants of the fit sentences) and re-ran the trust harness. On Brain-to-Text '25, 2 of 1,628 top-1 sentences changed and all main numbers were identical to three decimals. On sEMG, trust eWPM changed from 8.02, 4.68 and 4.08 to 7.98, 4.65 and 4.09 (paired participant CIs $[-0.08, -0.01]$, $[-0.05, -0.01]$, $[-0.01, +0.03]$), and no wrong-commit rate changed by more than 0.002. The re-tuned LM weights were unchanged.

H Datasets and licences

Table 9: Datasets, resources and licences.

Dataset / resource	Licence	Use	Note
SpanishBCBL [5, 82]	CC BY-NC 4.0	MEG decoder training and evaluation	Research only; weights trained on it are not released for commercial use
Brain-to-Text '25 copy-task logits [2, 34]	CC0 1.0	intracortical trust evaluation	Companion code has no licence; we wrote our own loader
Toronto sEMG typing [8, 81]	CC BY 4.0	sEMG decoder and trust evaluation	Baseline code (CC BY-SA 4.0) not used
Tatoeba English / Spanish sentences	CC BY 2.0 FR (some CC0)	character LMs; composed sEMG sentences	Sentences from Tatoeba (`https://tatoeba.org`), by the Tatoeba contributors
CMUdict (commit 7479086)	BSD-2-Clause style	phoneme-to-word lexicon	Copyright notice kept
Brain2Qwerty code (Meta)	CC BY-NC 4.0	read only, not copied	Our decoders are our own implementations
Fidetype code and specification	Apache-2.0	all methods	—

I MEG decoder details

The asynchronous MEG decoder is a convolutional front end (320 channels) followed by 4 Conformer layers of width 256 [83], trained with CTC [35] and a keystroke auxiliary loss that uses key times as training labels (key pretraining, then CTC training on all 80 recording blocks). At inference, the forward pass takes the MEG segment, its length and the participant index; it has no key input. Table 10 compares the inference inputs with the published asynchronous decoder. A longer training schedule gave the same validation CER and is not reported.

Table 10: Asynchronous setting: our decoder vs Brain2Qwerty v2.

Input at inference	Ours	v2 async [6]
MEG of the sentence segment	yes	yes
Segment start	first key − 0.4 s	first key − 400 ms
Segment end	last key press + 0.5 s	last key release + 400–500 ms
Per-key times or key count	no	no
Participant index	yes	yes
Key times as training labels	yes (auxiliary loss)	no

Leakage checks. The split is by a hash of the sentence text, and no test sentence occurs in the LM training text. A test sentence is "close" if a training sentence is within CER 0.25; 6 of the 17 test sentences are close. Without them, the CER was 0.519 (greedy) and 0.492 (beam).

Synchronous control decoder. A synchronous decoder with oracle key times (greedy CER 0.508, beam + LM 0.447; harness top-1 CER 0.451) gave the same trust results as the asynchronous decoder: always commit 2.714, trust 3.229 and gaze typing 3.155 eWPM, trust minus gaze [−0.02, +0.18] over sentences and [−0.07, +0.26] two-way. Over trials, this difference excluded 0 ([+0.03, +0.12]): trial-level intervals make the MEG result look like a gain, and sentence-level intervals do not.

J Format details and audit log

Serialisations and conformance. The reference serialisations of NeuralTextStream are an NPZ file and a JSON Lines streaming form. A conformance suite runs 60 checks, each tied to a specification clause, and reports one of three cumulative levels: *core* (a consumer must accept the stream), *trust-ready* (modality, causality, calibration state and language are declared) and *provenance* (stream id, decoder name, version and weights hash). It ships 52 golden test vectors (32 NPZ, 20 JSON Lines); the reference consumer passes 52 of 52. On synthetic inputs, our MEG and sEMG adapters reach the provenance level and the intracortical adapter reaches trust-ready, because the published online decoder has no weights file to hash. Writing the suite found five bugs in our own producers and ten ambiguities in the specification text, now listed as open questions.

Audit log. Each decision can be appended to a JSON Lines log. Context entries are content-addressed (their id is the SHA-256 of the context) and hold the adapter, decoder (name, version, weights hash, training data and licence), LM, calibration (temperatures, risk-control method, α, δ, threshold, n) and policy versions; decision entries hold the action, scores, a hash of the stream and its frame range; correction entries link to the decision they correct. Each entry stores the hash of the previous entry and its own SHA-256 over the canonical JSON, so any edit, deletion or reordering breaks the chain; a separately stored head anchor detects truncation. The log stores no raw signals (numeric arrays longer than 64 values are rejected), stores text as a keyed hash by default, and pseudonymises subject and session ids. It is tamper-evident, not tamper-proof: there is no signature in v0.

Replay. We replayed the Brain-to-Text '25 test half (816 trials) with the log attached, for the plug-in gate and a CRC gate. The log held 1,723 entries and passed verification. It recorded the same wrong commits as the harness: 48 of 48 (plug-in) and 39 of 39 (CRC), of which 2 and 1 were silent. This is a consistency check of the replay: the corrections come from the known cue sentences, and in live use silent errors would be missing from the log. Every tampering case we tried was detected: editing an entry, editing with re-hashing, deleting, reordering, and cutting the tail (with the head anchor). The cost was about 0.1 ms and about 1.1 kB per decision. Figure 7 shows one entry of the audit report.

Wrong commit `dec-000041-ed60b4bd` (entry 44, hash `8cf2c6c67053b185...`)
Committed: "who you never expect to when"
Correction (known cue sentence, hash-linked): "who you never expect to win"; 2 character edits, 1 word edit
Scores: $\hat{p}_{\text{top1}}$ 0.5814, set size 3, guard risk 0.2857; reason: $\hat{p}_{\text{top1}} \geq 0.50$
Input: stream `1b4b60e8e4950710...`, frames [0, 335], item `b2t25-test-trial-196`
Decoder: published online RNN + our phoneme-to-word decoder; LM: Tatoeba character 6-gram + word unigram + CMUdict; adapter `fidetype.adapters.b2t25`
Calibration `cal-692fb52a73417552`: τ_f 3.7809, τ_s 4.5551, fit set: first half of each day (812 trials)
Policy `pol-2eaa4ff51919527a`: commit threshold 0.5, maximum guard risk 0.5, offer size 2

Figure 7: One entry of the audit report (B2T '25 replay, plug-in gate; public CC0 cue sentence). Each wrong commit is traced to the exact stream segment and to the decoder, LM, calibration and policy versions that produced it. Hashes are shortened.

K Online re-calibration methods

Table 11: Online re-calibration methods for the commit threshold. L_t: realised loss of item t; T: number of items in the sequence; d: feedback delay in items. The two `aci` variants bound the long-run average of the *observed* losses for any sequence of items, without exchangeability; `window` and `decay` have no guarantee under shift.

Method	Update	Guarantee	Feedback
`window`	CRC on the last $N = 200$ items with known truth	none under shift[a]	full
`decay`	weighted CRC on all past items, weights ρ^{age}, $\rho = 0.99$	none under shift[a]	full
`aci`	$\lambda_{t+1} = \max(0, \lambda_t + \eta(L_t - \alpha))$, $\eta = 0.05$	$\frac{1}{T}\sum_t L_t \le \alpha + \frac{\lambda_{T+1} - \lambda_1}{\eta T}$ [b]	realised only
`aci_window`	window CRC at level α_t; $\alpha_{t+1} = \min(2\alpha, \alpha_t + \gamma(\alpha - L_t))$, $\gamma = 0.005$	$\frac{1}{T}\sum_t L_t \le \alpha + \frac{\alpha_1 - \alpha_{T+1}}{\gamma T}$ [c]	full + realised

[a] The weighted-CRC bound $\alpha + \sum_i \tilde{w}_i \, d_{\text{TV}}(Z, Z^{(i)})$ [58, 59] cannot be computed. Items whose truth is unknown are left out of the window, which can bias it.
[b] The right-hand side is a certificate computed after the fact. It is bounded in advance only if λ_t stays bounded, which holds for scores ≤ 1 and $f \equiv 0$; then $\frac{1}{T}\sum_t L_t \le \alpha + \frac{1 - \lambda_1 + \eta(1+d)}{\eta T}$.
[c] The upper clip at 2α only lowers α_t, so the bound stays valid. α_t stays bounded below because below the CRC feasibility level ($\approx 1/(N+1)$) nothing is committed and α_t rises.
All bounds count only the losses that the controller observes. Items whose feedback has not arrived are skipped; in our runs all feedback arrived after the delay, except in the "noticed errors only" runs, where a silent error is fed back as a correct commit.

L How to state the guarantees

With CRC: "if the calibration sentences and later sentences are exchangeable, and the calibration sentences fitted no other part of the system, then, averaged over the choice of the n calibration sentences and over a new sentence, the probability that the new sentence is committed wrongly is at most 0.05". A single session can exceed 0.05. With LTT: "under the same assumptions, with probability at least 0.9 over the calibration sentences, the long-run wrong-commit rate at the chosen threshold is at most 0.05". With adaptive online re-calibration: "for any sequence of sentences, the average observed wrong-commit rate over the first T sentences is at most $0.05 + (1 - \lambda_1 + \eta)/(\eta T)$". All three count only the errors that the system observes, so each should be reported with the measured silent-error rate, and none of them says anything about the error among commits.

M CRC and LTT on random splits

Table 12 gives CRC and LTT at four calibration sizes (500 random splits). Figure 8 shows the spread of the test wrong-commit rate over 200 random calibration/test splits of the test pool for 100 and 408 calibration items (the splits behind the pool-risk numbers in Section 7.6). The fixed gate is the plug-in policy fitted on the fit data.

Table 12: CRC and LTT on random calibration/test splits of the test pool ($\alpha = 0.05$, $\delta = 0.1$; 500 splits). Wrong-commit: mean test rate; p90: its 90th percentile; $P_{>\alpha}$: share of splits with a test rate above α; Commit: mean commit rate; Err. | commit: pooled over splits; Never: share of splits without a commit. MEG: pool of 637 instances.

Data	Method, n_{cal}	Wrong-commit	p90	$P_{>\alpha}$	Commit	Err. | commit	Never
B2T '25, our decoder	CRC, 20	0.0308	0.089	0.22	0.123	0.25	0.17
	CRC, 50	0.0342	0.071	0.21	0.146	0.23	0
	CRC, 100	0.0456	0.075	0.38	0.186	0.25	0
	CRC, 408	0.0479	0.069	0.43	0.192	0.25	0
	LTT, 20	0	0	0	0	—	1.00
	LTT, 50	0.0201	0.046	0.09	0.097	0.21	0
	LTT, 408	0.0356	0.052	0.15	0.156	0.23	0
B2T '25, published gate	CRC, 20	0.0308	0.073	0.23	0.345	0.09	0.03
	CRC, 408	0.0495	0.071	0.46	0.484	0.10	0
	LTT, 408	0.0355	0.054	0.15	0.429	0.08	0
MEG, asynchronous	CRC, 408	0.0197	0.031	0	0.021	0.93	0
	LTT, 408	0.0220	0.031	0	0.024	0.93	0

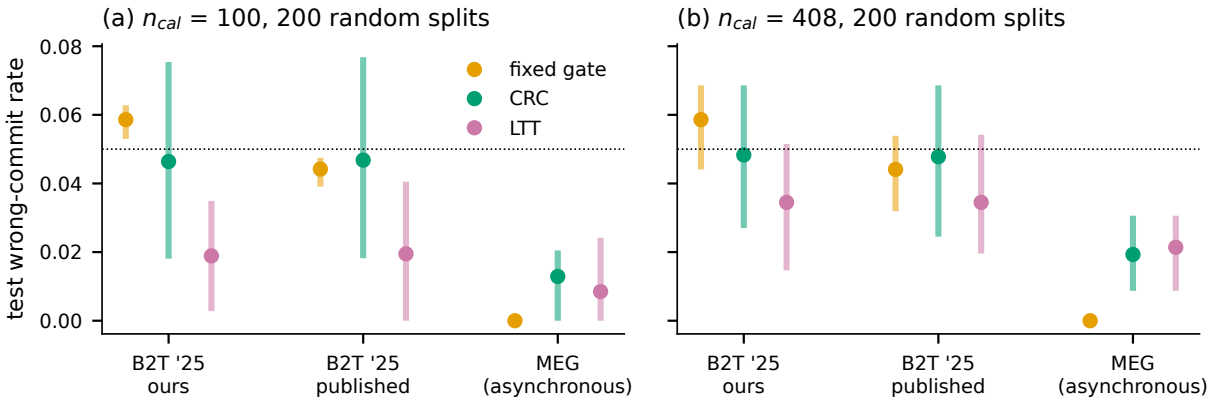


Figure 8: CRC bounds the mean over splits; LTT bounds the calibration-conditional risk with probability $1 - \delta$. Test wrong-commit rate over 200 random calibration/test splits with (a) 100 and (b) 408 calibration items. Dots: mean; bars: 5th to 90th percentile (the 90th percentile matches $\delta = 0.1$). Dotted line: $\alpha = 0.05$. A test rate above α in a single split is not a violation of either guarantee.